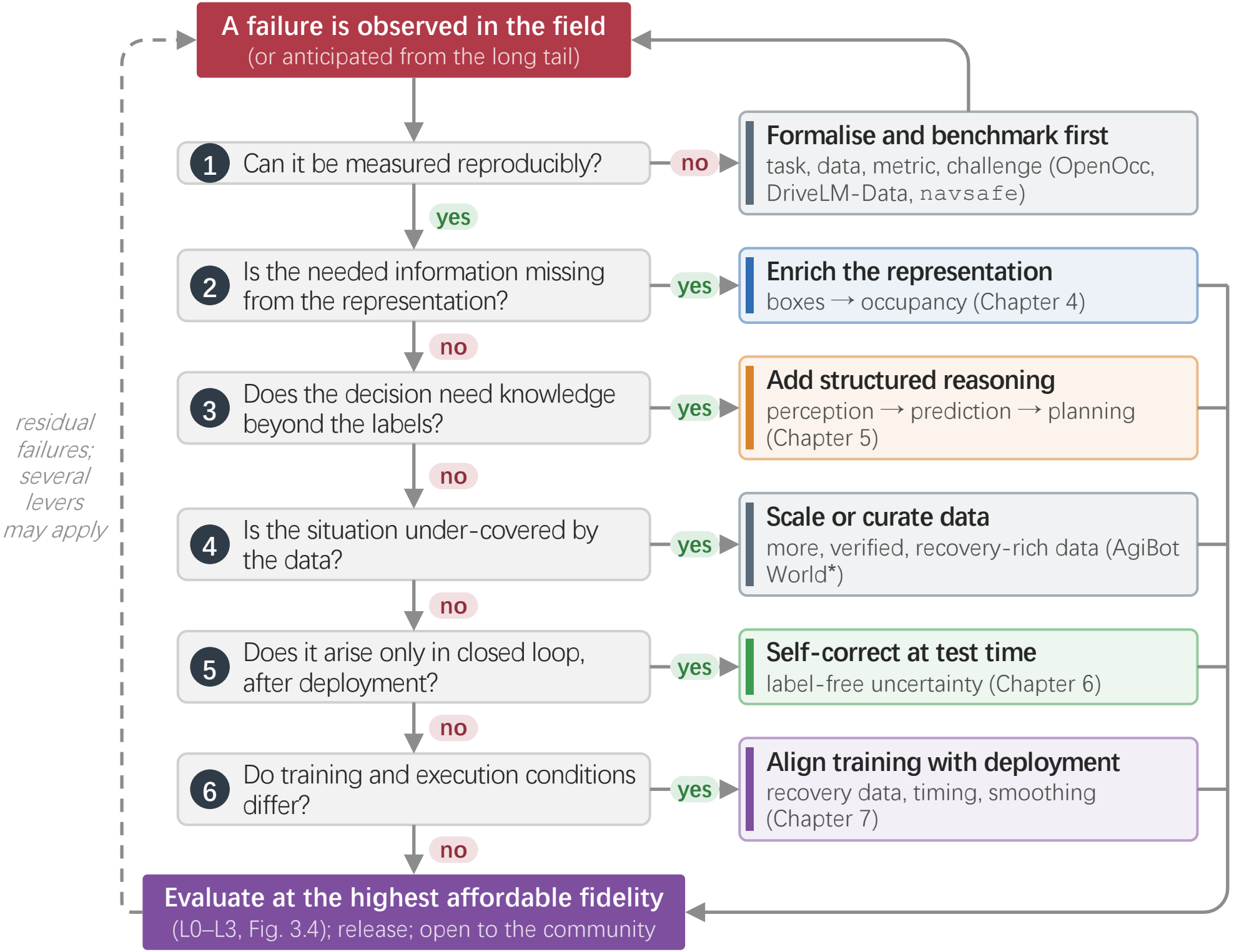
A failure is observed in the field
(or anticipated from the long tail)
1 Can it be measured reproducibly?
no
Formalise and benchmark first
task, data, metric, challenge (OpenOcc, DriveLM-Data, navsafe)
yes
2 Is the needed information missing from the representation?
yes
Enrich the representation
boxes → occupancy (Chapter 4)
no
3 Does the decision need knowledge beyond the labels?
yes
Add structured reasoning
perception → prediction → planning (Chapter 5)
no
4 Is the situation under-covered by the data?
yes
Scale or curate data
more, verified, recovery-rich data (AgiBot World*)
no
5 Does it arise only in closed loop, after deployment?
yes
Self-correct at test time
label-free uncertainty (Chapter 6)
no
6 Do training and execution conditions differ?
yes
Align training with deployment
recovery data, timing, smoothing (Chapter 7)
no
Evaluate at the highest affordable fidelity
(L0–L3, Fig. 3.4); release; open to the community
residual failures; several levers may apply